HAI-Agency: Workshop on Orchestrating Human and AI Agency for Proactive and Reflective Learning

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Abstract. As research momentum shifts toward the next evolution of generative AI (GenAI), Agentic AI, educational technologies are moving beyond reactive tools toward *proactive, teammate-like ecosystems* grounded in pedagogical principles. This transition raises a central challenge: how to design increasingly autonomous AI systems without diminishing learner agency or undermining teachers' professional judgement. This workshop introduces the concept of **HAI-Agency**, envisioning how human and AI agency can be orchestrated in learning and teaching. Foregrounding proactive and reflective learning, we aim to advance a shared research agenda spanning design methodologies, computational modeling, evaluation frameworks, and the classroom integration of agentic AI systems. This workshop is of particular interest to the AIED community, especially researchers and practitioners across human-AI interaction, learning analytics, AI-supported learning design, and learning sciences. Through an interactive, hybrid, half-day format, we will explore how to shape a future in which agentic AI contributes to more equitable, transparent, and pedagogically sound educational innovations.

Keywords. Proactive learning, Learner agency, Agentic AI, Human-AI interaction, AI-supported learning design

1 Introduction

Recent advances in large language models (LLMs) and multimodal generative AI (GenAI) have transformed educational systems across many areas, including automated feedback, content generation, and adaptive tutoring. However, two gaps are limiting educational impact and motivating a new research agenda. First, most current GenAI applications remain reactive, relying on explicit prompts and continuous human guidance rather than autonomous, goal-directed support (Lee & Palmer, 2025). The AIED community also recognizes that GenAI is no longer just a "reactive tool" but is evolving toward "proactive teammate-like ecosystems". Second, GenAI adoption has outpaced the adoption of pedagogical design principles and

evidence standards (Kostopoulos et al., 2025). The OECD emphasizes that GenAI can improve *task performance* without necessarily producing *learning gains* unless it is guided by clear pedagogical principles and designed to strengthen durable cognitive and metacognitive skills (OECD, 2026). In response, attention is shifting to the next evolution of GenAI: **Agentic AI**, a goal-directed AI system that not only responds but also perceives context, plans, and acts to pursue learning and teaching objectives.

This creates a core tension for human-AI collaboration in education: **increasing AI autonomy must not reduce learner agency or undermine teacher professional judgement**. Education succeeds when learners and teachers take initiative to set goals, monitor progress, reflect, and orchestrate complex social and classroom realities. As we move from single-chat interactions to multi-agent, co-adaptive systems, pedagogical alignment becomes more challenging: pedagogical principles must remain stable across agents, systems, and time while still preserving learner and teacher agency.

To address the technological and pedagogical gaps, we must rethink **design** (agent autonomy vs. human agency), **modeling** (human-AI co-orchestration, shared goals and initiative, pedagogical alignment), **evaluation** (new learning analytics that can distinguish overreliance from productive co-learning), and **integration** (fit with classroom practices, teacher workflows, and institutional governance). With AIED 2026 explicitly calling for a shift "from tools to teammates," this is the right moment for a community-level discussion that connects AIED with the wider edtech ecosystem. The workshop will therefore **network** researchers, practitioners, and policymakers participating in AIED 2026 and the Festival of Learning to address these gaps through an interactive, hybrid, half-day format.

We expect three **main outcomes** from the workshop. First, a community blueprint for Agentic AI in Education that synthesizes shared definitions, core design principles, pedagogical-alignment policies, and guidance for responsible deployment. Second, peer-reviewed papers in the workshop proceedings that capture diverse perspectives, concepts, prototypes, and early empirical findings. Third, a collaborative research agenda that prioritizes future work by articulating key research questions, methodological needs, and evidence standards for evaluation and classroom integration.

2 Relevance and Importance to the AIED Community

This workshop directly addresses the **AIED 2026 theme** "From Tools to Teammates: Human–AI Synergy for Augmented Learning" by focusing on agentic AI, the shift from AI as passive tools into active collaborators in education. It targets a core **scientific question** at the heart of this transformation: *when multi-agentic systems become proactive and autonomous, how can we preserve or strengthen learner and teacher agency rather than replacing it?* Answering this is not only a technological challenge, but a defining AIED challenge: designing educational "teammates" that are pedagogically aligned, trustworthy, and deployable in real learning ecosystems.

The workshop will help the community define foundational concepts and methods for **agentic AI in education**. Through structured discussion and a hands-on demonstration, participants will co-produce concrete community outputs, including

robust metrics and analytics for agency and co-regulation, criteria for pedagogical alignment, and a prioritized set of research questions and validation needs that can inform future AIED/EDM/LA work. This aligns with the Festival of Learning's emphasis on workshops that connect communities with clear needs and outcomes, and it advances the **multidisciplinary integration** AIED 2026 calls for by linking technical agent architectures with learning-science-grounded design and evidence.

3 Content and Themes

The call for participation will welcome theoretical or empirical submissions of position papers, case studies, or ongoing research on the following topics:

- **Learning goals and pedagogical foundations**:
  Proactive/reflective learning, self-regulated learning/co-regulation/learner agency, affective/cognitive/behavioral engagement, motivational orientations, competencies for the GenAI era (e.g., creativity, higher-order thinking, critical thinking)
- **Interaction, intervention & learning design**:
  Proactive/reflective prompts, scaffolds, and interventions, learning analytics-driven feedback and recommendations, teacher-facing design and support, human-centered and personalized interactions, creative AI-enhanced learning, human-computer interactions
- **Modeling & analytics**:
  Student/teacher/context modeling, human-AI interaction modeling, learning sequence analysis and process mining
- **Evaluation & assessment**:
  Process-based assessment, outcome and process trade-offs, automated assessment, human-AI collaborative assessment, integrity-aware evaluation, assessment reinvention of GenAI-assisted competencies and AI-powdered skills
- **Agentic human-AI orchestration**:
  Agency/automation, negotiation and coordination, explainable/teachable/designable AI
- **Ethics and social impact**:
  Safety/integrity/fairness/robustness/uncertainty/calibration/governance, challenges in classroom-level deployment, social infrastructure upgrading with AI agents, human-AI symbiosis

4 Expected Target Audience

This workshop is designed for a broad audience within the AIED community, including policy experts, researchers, practitioners, teachers, and students with diverse levels of expertise. We welcome both participants new to the topic and experienced scholars working in related areas, such as human-AI interaction, learning analytics,

AI-supported learning design, and the learning sciences. To support inclusiveness and broad participation, the workshop will be held in a hybrid format (on-site and online), enabling engagement from participants across different regions and time zones. We particularly encourage participation from interdisciplinary researchers and practitioners interested in proactive and reflective learning, human-AI collaboration, and agentic AI in education. We expect approximately 20–30 participants. Participation will be open: attendees may join the workshop discussions without submitting a paper or giving a presentation. This format is intended to foster active exchange and community building among participants from diverse backgrounds.

We will advertise the workshop across multiple channels to reach a broad, diverse audience. In particular, we will disseminate the call for papers and participation through our professional networks, relevant mailing lists, and academic communities. We will also share the workshop announcement via social media (e.g., X, LinkedIn) and institutional websites where appropriate. In addition, we will directly invite researchers and practitioners whose work aligns with the workshop themes to encourage participation from both established scholars and emerging researchers.

5 Format and Activities

We propose a half-day workshop with a keynote talk, two presentation sessions, a hands-on demo, and group discussions in the afternoon. Online participants will also be supported if required. We have provided a detailed tentative schedule in Table 1.

Table 1. Schedule of activities

Time	Activity
14:00 - 14:10	**Opening Remarks** Welcome and overview of the workshop goals, theme, and expected outcomes. The organizers will introduce the concept of *HAI-Agency*, outline the central tension between AI autonomy and human agency in education, and frame the key questions that will guide the day.
14:10 - 14:30	**Keynote Talk** Invited keynote on the evolution from reactive GenAI tools to agentic AI teammates in education. The talk will examine how proactive, goal-directed AI systems can be pedagogically aligned while preserving learner and teacher agency. The keynote will provide conceptual grounding, real-world examples, and future research directions for human-AI orchestration in learning.

14:30 - 15:30	**Presentation Session 1** (4-6 talks) Research presentations focus on theoretical foundations, modeling approaches, and early empirical work related to learner agency, proactive AI interventions, and human-AI collaboration. A guided thematic discussion will be followed by the presentations.
15:30 - 15:45	Coffee Break
15:45 - 16:45	**Presentation Session 2** (4-6 talks) Research presentations center on design practices, evaluation frameworks, classroom integration, ethics, and deployment challenges of agentic AI systems. Discussions will emphasize methodological rigor, evidence standards, and implications for real-world educational contexts.
16:45 - 17:15	**Hands-on Demo** Interactive demonstration of agentic AI concepts in practice, including example workflows using LLMs, agent architectures, tools, and orchestration frameworks such as Agent Development Kit (ADK) by Google, Model Context Protocol (MCP) servers, etc. Participants will explore how proactive prompts, multi-agent coordination, or reflective scaffolds can be implemented in a sample educational scenario (e.g., mathematics learning). The session aims to bridge theory and applied design.
17:15 - 17:45	**Group Discussion** (Opportunities and Challenges) Structured small-group and plenary discussion on key opportunities and risks of integrating agentic AI into educational ecosystems. Participants will collaboratively identify priority research questions, design principles for preserving human agency, evaluation criteria, and governance considerations. Outcomes will contribute to the community blueprint for Agentic AI in Education.
17:45 - 18:00	**Wrap Up and Closing**

6 Workshop Timeline

The plan to implement the workshop, including the CFP, spans three months before the conference and the month after. This is to allow sufficient time for the authors to submit their contributions and manage the conference registration on time.

Important dates (AOE):

– Publishing workshop's website: March 22, 2026

– Launch call for papers: March 22, 2026
– Paper submission deadline: April 15, 2026
– Paper reviews: from April 16 to May 6, 2026
– Notification of acceptance: May 11, 2026
– Paper camera-ready version: May 25, 2026
– Workshop: June 28 or June 29, 2026
– Workshop proceedings submission to CEUR: July 31, 2026

7 Plan for Publishing the Workshop's Proceedings

This workshop will issue a call for papers inviting submissions of full papers (8–10 pages), short papers (5–6 pages), and position papers (2–3 pages). All page limits include references. A detailed list of subtopics will be announced to help prospective authors ensure their submissions align with the workshop's scope. All submissions will undergo a peer-review process, with at least 2 reviewers per paper. Papers will be selected based on criteria including quality, rigor, originality, relevance to the workshop theme, and consideration of ethical and societal implications.

To ensure the visibility and accessibility of accepted contributions, the workshop chairs will compile the proceedings and submit them for publication in CEUR Workshop Proceedings, drawing on our prior experience with this venue (Flanagan et al., 2022).

8 Organizers and Advisors

The organizers and advisors are collaborators with interdisciplinary research backgrounds in AIED, EDM, LA, and HCI. Over the past several years, the organizers have conducted joint workshops at various international conferences and joint projects in AIED (Ma et al., 2025; Ogata et al., 2024; Yang et al., 2021). Those workshops spanned topics such as Learning Analytics Data Challenges at LAK (2018-2025), Learning Analytics workshops at ICCE (2018-2022), the GenAI in Education workshop at ICCE (2024), and the MiXai^learn workshop at AIED (2025).

8.1 Workshop Chairs

Dr. Yiling Dai is an Assistant Professor at the Graduate School of Advanced Science and Engineering, Hiroshima University, Japan. She has led and co-led four national research projects in Japan and serves as an active program committee member for leading international conferences (LAK, ICCE, and AIED), being awarded the best reviewer award at ICCE2024. Her research interests include knowledge representation, educational data mining, learning analytics, and educational recommendations, with a special focus on explainable AI-driven recommendations and interventions.

Dr. Boxuan Ma is an Assistant Professor in the Faculty of Arts and Science at Kyushu University and an adjunct member of the School of Interdisciplinary Science and Innovation at Kyushu University. His research lies at the intersection of Artificial Intelligence in Education, Educational Data Mining, Learning Analytics,

Human-Computer Interaction, and Recommender Systems. He serves on the program committees of several leading international conferences in the field, including LAK, EDM, and AIED.

Dr. Huiyong Li is an Assistant Professor at the Research Institute for Information Technology, Kyushu University, Japan. His research focuses on learning analytics, educational data mining, and AI in education. He has led or co-led four national educational technology projects in Japan and serves as a Virtual Experience Co-Chair for AIED 2026. His work has received national and international recognition, including an Outstanding Research Award (Kyoto University) and the Best Short Paper Award (ICLEA 2025).

Dr. Patrick Ocheja is an AI, Data, and Cloud practitioner based in Canada and a former JSPS Postdoctoral Fellow at the Academic Center for Computing and Media Studies, Kyoto University, where he also earned his Ph.D. in Informatics. His research and professional work span Artificial Intelligence in Education (AIED), Learning Analytics, personalized and lifelong learning systems, and the design of scalable AI-driven infrastructure. He has previously served as a program committee member and workshop organizer at ICCE and LAK, and co-organized APSCE Student Wing activities from 2019 to 2023, serving as co-chair in 2021 and chair in 2022.

Prof. Kyoungwon Seo is an Associate Professor in the Department of Applied Artificial Intelligence at Seoul National University of Science and Technology, South Korea. His research focuses on Human-Centered AI, Artificial Intelligence, and intelligent interactive systems, integrating multimodal analytics, large language models, and multi-agent architectures to enhance human cognition and learning. He has led nationally funded research projects and interdisciplinary collaborations with industry partners, including Microsoft, Hyundai Motor Group, and healthcare organizations. His work has received international and national recognition, including the 2024 High Impact Award from the International Journal of Educational Technology in Higher Education and the 2023 Minister's Commendation from the Ministry of Trade, Industry and Energy (South Korea).

Prof. Brendan Flanagan is an Associate Professor at the Center for Innovative Research and Education in Data Science, Institute for Liberal Arts and Sciences, and the Graduate School of Informatics at Kyoto University. His research spans across the fields of Artificial Intelligence in Education (AIED), Learning Analytics, Educational Data Science, Educational Data Mining, NLP/Text Mining, and Computer-Assisted Language Learning. He is the principal investigator on three national-level research projects and also works as part of a Japanese government-funded large research project into educational symbiotic AI systems. He hosted and chaired LAK2024 in Kyoto and has organized workshops at LAK, AIED, and ICCE.

8.2 Workshop Advisors

Prof. Hiroaki Ogata is a Professor and Director at the Academic Center for Computing and Media Studies (ACCMS), Kyoto University, Japan. He has published more than 700 peer-reviewed papers in SSCI Journals and top international conferences. He has received APSCE (Asia-Pacific Society for Computers in Education) Distinguished Researcher Award in 2014, and several Best Paper Awards, and has given keynote lectures in several countries. Currently, he is the President of

APSCE, an associate member of the Science Council of Japan, and the Director of Evidence-Driven Education Research Council, Japan.

Prof. Stephen J.H. Yang is Chair Professor of Computer Science and Information Engineering at National Central University and was the first Director of the Department of Information and Technology Education at the Ministry of Education, Taiwan. He is a world-leading computer scientist and education scholar with over 100 journal articles, more than 16,000 Google Scholar citations, and an h-index of 56. His awards include Taiwan MOE's 2025 Excellent Teacher Award, the 69th Academic Award, and two National Science Council Outstanding Research Awards.

Prof. H. Ulrich Hoppe is an emeritus professor of the University of Duisburg-Essen (Germany). He is a fellow of the Asia-Pacific Society for Computers in Education (APSCE) and of the International Society of the Learning Sciences (ISLS). His research has focused on computer support for learning and knowledge building in various contexts, including higher education and vocational education and training. With his COLLIDE research group, founded in 1995, he has participated in more than ten EU projects on Technology-enhanced Learning with a special focus on intelligent, AI-based support for collaborative learning and scientific inquiry.

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